

MÖBIUS CANCELLATION IN ALL INTERVALS OF LENGTH $x^{0.5435}$

AUTOMATED MULTI-AGENT PROOF SEARCH (CLAUDE, ANTHROPIC)

ABSTRACT. We give an argument, partly computer-assisted and refereed only internally, that for every $\vartheta > 0.5435$ and $\varepsilon_0 > 0$ one has $\sum_{x < n \leq x+H} \mu(n) \ll H(\log x)^{-1/3+\varepsilon_0}$ for all $H \geq x^\vartheta$. This would lower the exponent 0.55 of Matomäki and Teräväinen by 0.0065. Their reduction is kept unchanged. The only new step replaces their Heath-Brown–Iwaniec type lemma. In its place we use a level-set argument built on the Guth–Maynard large values estimates, including the $\inf_k$ form of their Proposition 12.1, and on the fourth moment of ζ . A branch-and-bound certificate in exact rational arithmetic, with margin 10^{-5} , closes the exponent bookkeeping. The binding configuration is five smooth factors of length $x^{1/5}$. With our inputs it closes exactly when $\vartheta > 69/127 = 0.5433071\dots$, which is the limit of the method. The certificate also passes at $\vartheta = 0.543308 = 69/127 + 9.1 \cdot 10^{-7}$, and a control run at $\vartheta = 0.5433$ fails. We also record why the routes tried in this project do not go below $69/127$; in the binding case the problem is a fifth-moment bound for a single segment of the zeta function.

1. INTRODUCTION

Let μ be the Möbius function. Matomäki and Teräväinen [MT, Theorem 1.1] proved the following. *Let $\theta > 0.55$ and $\varepsilon > 0$ be fixed. Then for x large enough and $H \geq x^\theta$, $\sum_{x < n \leq x+H} \mu(n) = O(H/(\log x)^{1/3-\varepsilon})$.* The exponent $0.55 = 11/20$ is the Heath-Brown–Iwaniec exponent for primes in short intervals [HBI]. It enters [MT] through one lemma, [MT, Lemma 4.4], which they take from [HBI, Lemma 2] (see also [Ha, Lemma 10.12]). MT remark in [MT, §2.1] that the proof of that lemma is not continuous in θ and depends essentially on $\theta > 0.55$. This is a paraphrase of text obtained through a summarising fetch tool. In the level-set model used below, the classical large values inputs (the mean value theorem, Huxley’s estimate [Hu] and Jutila’s estimate [Ju]) are consistent with this exponent: the same code certifies 0.552 and fails at 0.548 (§6). Guth and Maynard [GM] have since improved large values estimates for Dirichlet polynomials taking values near $N^{3/4}$. This note records what their estimates give in the Matomäki–Teräväinen framework. An earlier version of this note proved the result below for $\vartheta > 0.548$, using only the closed form of [GM, Proposition 12.1].

Theorem 1.1. *Let $\vartheta > 0.5435$ and $\varepsilon_0 > 0$ be fixed. Then for x large enough and $H \geq x^\vartheta$,*

$$\sum_{x < n \leq x+H} \mu(n) \ll \frac{H}{(\log x)^{1/3-\varepsilon_0}}.$$

In particular, the exponent $11/20$ of [MT] can be lowered by 0.0065: for every fixed $\vartheta > 11/20 - 0.0065$ one has $\sum_{x < n \leq x+H} \mu(n) = o(H)$ for all $H \geq x^\vartheta$.

Status. The proof (§§2–5) is complete as written, modulo three things.

- (a) The cited results. The statements from [MT] were read only through a summarising tool. An independent re-check (S3), also via such a tool, agrees (Remark 3.8). Classical results are cited as standard, without theorem numbers.
- (b) A computer certificate (§4.7) in exact rational arithmetic, with margin 10^{-5} . It extends the certificate of the 0.548 version (run in double precision and ported to exact arithmetic by a separate agent) by one estimate, the $\inf_k$ form of [GM, Proposition 12.1]. The extension was written by one agent (GA). The referee R4 re-ran it, and re-checked every accepted leaf with an

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independently written script (§4.7); the writer of this note also re-ran the headline invocation and obtained the same tiling.

- (c) Refereeing. The proofs of the 0.548 version were refereed by an independent agent (R2), with verdict *correct with minor fixable issues*; all issues were addressed. A second, fresh-eyes referee (R3) read that note; its minor issues were also applied. The changes made for 0.5435 (listed in §7) were refereed by a third independent agent (R4), with verdict *accept*: no blocking or major issues, and four minor ones, all addressed here. There has been no human refereeing.

We regard the result as *unverified* until it has been checked outside this project.

Method. We follow [MT] verbatim up to the point where [MT, Lemma 4.4] is used (§2). In its place we prove Lemma 4.1 (§4). The configuration is a product ABC of Dirichlet polynomials: A and B are arbitrary and divisor-bounded, and C is a smooth (zeta-type) partial sum. We need an L^1 bound for ABC on $[N_C, x^\tau]$ with $\tau \approx 0.4565$. We discretise and split into level sets. On each level set the number of points is bounded by the smallest of several estimates:

- the mean value theorem;
- Guth–Maynard’s Theorem 1.1 and the $\inf_k$ form of their Proposition 12.1, applied to a fixed finite family of products of A, B, C ;
- a fourth-moment bound for C , which we transfer from ζ (Lemma 4.3).

Three regions need separate arguments. Small t and long C are handled by Cauchy–Schwarz. The “corner” where $|A|$ and $|B|$ are both nearly maximal is handled by the Halász–Montgomery inequality and the Vinogradov–Korobov saving of the prime polynomial. A rigorous box certificate covers the rest of parameter space.

Limits. With smaller margins the same certificate passes at $\vartheta = 0.543308 = 69/127 + 9.1 \cdot 10^{-7}$ (Remark 5.1). The number $69/127 = 0.5433071\dots$ is the limit of the method. For five smooth factors of length $x^{1/5}$ the inputs used close the case exactly when $\vartheta > 69/127$; this is a refereed computation (P2, R1), recomputed exactly in phase II (PB1). A control run of the certificate at $\vartheta = 0.5433$ fails. Section 6 records what phase II of the search found about going below $69/127$. In the binding case the problem is a fifth-moment bound for one zeta segment. A joint large values route is refuted. An additive-energy input inside the Guth–Maynard method could gain at most $69/127 - 19/35 \approx 0.00045$. We also list priced targets. Finally, §6.1 records a refereed reduction below $69/127$. For $\vartheta > 0.542$, everything outside a near-equal family of configurations is certified.

2. THE MATOMÄKI–TERÄVÄINEN REDUCTION

Throughout, $T_0 = \exp((\log x)^{1/3})$. Write $f \ll_A g$ when the constant depends on A ; all other constants may depend on the fixed parameters. We use MT’s notation with a general exponent. Fix $\theta \in (0.53, 0.56)$ and a small $\varepsilon > 0$. Let $H \asymp x^{\theta+\varepsilon}$. Take $P = \exp((\log x)^{2/3+\varepsilon/2})$, $Q = x^{1/(\log \log x)^2}$ and $k = 20$, as in [MT, §4.1]. MT take $\theta = 0.55$.

Lemma 2.1 ([MT, Lemma 3.1]). *Let $\varepsilon > 0$ be fixed, x large, $(\log x)^4 \leq P < Q \leq x^{o(1/\log \log x)}$ and $x^\varepsilon \leq H \leq x$. Then*

$$\sum_{x < n \leq x+H} \mu(n) = - \sum_{\substack{x < prn \leq x+H \\ P < p \leq Q, \ r \leq x^{\varepsilon/2}}} a_r \mu(n) + O\left(H \frac{\log P}{\log Q}\right),$$

with explicit divisor-bounded coefficients a_r (a Ramaré-type weight).

With the above P, Q , the error term is $\ll H(\log x)^{-1/3+\varepsilon}$. This is the only source of the exponent $1/3$. Heath-Brown’s identity with $k = 20$ and a dyadic decomposition [MT, §4.1] reduce matters to $\ll (\log x)^{2k+2}$ sums over $x < prn_1 \cdots n_{2k-1} \leq x+H$, with $p \sim P_1, r \sim R$ and $n_i \sim N_i$. The parameters satisfy

$$(1) \quad P_1 \in [P, Q], \quad R \in [\tfrac{1}{2}, x^{\varepsilon/2}], \quad N_i \in [\tfrac{1}{2}, x] \ (i \leq k-1), \quad N_i \in [\tfrac{1}{2}, (2x)^{1/k}] \ (i \geq k), \quad P_1 R \prod_i N_i \asymp x.$$

For $i \leq k-1$ the coefficients $a_i(n)$ are 1 or $\log n$; we call these factors *smooth*. For $i \geq k$ they are of Möbius type, $\mu(n)\mathbf{1}_{n \leq (2x)^{1/k}}$. Each such sum must be compared with H/y_1 times the same sum over $(x, x+y_1]$, where $y_1 = x \exp(-3(\log x)^{1/3})$; call this comparison $(*)$. Write $P(s) = \sum_{P_1 < p \leq 2P_1} p^{-s}$, $R(s) = \sum_{r \sim R} a_r r^{-s}$ and $N_i(s) = \sum_{n \sim N_i} a_i(n) n^{-s}$.

Lemma 2.2 ([MT, Lemma 4.2]). *Suppose that for all $A > 0$,*

$$(2) \quad \int_{T_0}^{x^{1+\varepsilon/10}/H} |PRN_1 \cdots N_{2k-1}(\tfrac{1}{2} + it)| dt \ll_A x^{1/2} (\log x)^{-A}.$$

Then $()$ holds with error $O_A(H(\log x)^{-A})$.*

Lemma 2.3 (Vinogradov–Korobov; proof of [MT, Lemma 4.3]). *For $|t| \leq x^2$, $|P(1+it)| \ll P_1^{-1/(\log x)^{2/3+\varepsilon/3}} (\log x)^3 + \log x/(|t|+1)$. The same bound holds for every partial sum over $(P_1, u]$ with $u \leq 2P_1$. Partial summation then gives $\sup_{T_0 \leq |t| \leq x^2} |P(\tfrac{1}{2} + it)| \ll_A P_1^{1/2} (\log x)^{-A}$. This is the step MT use in their proof of [MT, Lemma 4.3], whose final bound carries the factor $P_1^{1/2} (\log x)^{-A}$.*

Lemma 2.4 (Type II; [MT, Lemma 4.3] with a general window). *Let $T = x^{1+\varepsilon/10}/H$ and $X_I = \prod_{i \in I} N_i$. If some $I \subset [2k-1]$ has both $X_I \geq T$ and $X_{[2k-1] \setminus I} \geq T$, then (2) holds.*

Proof. This is MT's proof. Lemma 2.3 bounds P and the trivial bound handles R . The rest is Cauchy–Schwarz and the mean value theorem, Lemma 3.1: $\int_{T_0}^T |\prod_I N_i \prod_{I^c} N_i| \ll (T + X_I)^{1/2} (T + X_{I^c})^{1/2} (\log x)^{O(1)} \ll (X_I X_{I^c})^{1/2} (\log x)^{O(1)}$. $\square$

MT state the window as $X_I \in [x^{0.45-\varepsilon/2}, x^{0.55+\varepsilon/2}]$. At its upper end the complement condition holds only after a harmless adjustment: either group R with I^c , or note that MT's Lemma 4.5 only produces $X_I \in [x^{0.45}, x^{0.55}]$.

Lemma 2.5 (Combinatorics; cf. proof of [MT, Lemma 4.5]). *Let $0 \leq w_1 < w_2$, $\alpha_i > 0$ and $S = \sum_i \alpha_i > w_2$. Suppose no subsum lies in $[w_1, w_2]$. Let I have the largest subsum $\leq w_2$. Then $\sum_I \alpha_i < w_1$. Moreover, some $r \notin I$ exists, and every $r \notin I$ satisfies $\alpha_r > w_2 - \sum_I \alpha_i$ and $\sum_J \alpha_j < S - w_2$, where $J = [K] \setminus (I \cup \{r\})$.*

Proof. $\sum_I \leq w_2$ is not in $[w_1, w_2]$, so it is $< w_1$. Since $\sum_I < S$, some $r \notin I$ exists. Maximality and $\alpha_r > 0$ give $\sum_I + \alpha_r > w_2$, and then $\sum_J = S - \sum_I - \alpha_r < S - w_2$. $\square$

Remark 2.6 (θ -independence of the rest of MT's proof; audit). In the proof of [MT, Theorem 1.1], the hypothesis $\theta > 0.55$ enters only in five places (lemma numbers refer to [MT]):

- (i) Lemma 4.4;
- (ii) the window of Lemma 4.3, which comes from the cut $T = x^{1+\varepsilon/10}/H$;
- (iii) the numbers in Lemma 4.5;
- (iv) the type I Lemma 4.1, which MT note is already covered by the other two lemmas (paraphrase);
- (v) the check that $k = 20$ and $N_r \gg x^{0.079}$ in the proof of Lemma 4.4.

Everything else uses only $x^\varepsilon \leq H \leq x$:

- Lemma 2.1;
- Heath-Brown's identity and the reduction to $H \asymp x^{\theta+\varepsilon}$;
- Lemma 2.2, including the ranges $|t| \leq T_0$ and the polynomial R ;
- Lemma 2.3;
- the final step through the prime number theorem with Vinogradov–Korobov error term.

This is the conclusion of an audit of [MT] (internal report S2). An independent re-check of the MT text (S3) agrees that only [MT, Lemma 4.4] needs $\theta > 0.55$; its quotes were also obtained through a summarising web tool. The S2 audit used the arXiv PDF and the ar5iv rendering, read through a summarising fetch tool. For Lemma 2.2, MT cite [Ha, Lemma 7.2], which we could not consult. The

audit instead re-derived the Perron step and found it uniform in θ . So, by Remark 2.6, items (i)–(v) are the only changes needed. Lemma 2.4 and Lemma 2.5 take care of (ii)–(iv). Item (v) holds because $2\theta - 1 - \varepsilon > 1/20$ for $\theta > 0.53$. Item (i) is Lemma 4.1 below.

3. ANALYTIC INPUTS

For each input we give the statement in the form used. “ $\lesssim$ ” means $\leq C_\eta T^\eta$ for every $\eta > 0$.

Lemma 3.1 (Mean value theorem; standard, see [MV], [Ga]). $\int_0^U |\sum_{n \leq N} a_n n^{-it}|^2 dt = \sum |a_n|^2 (U + O(n))$. Also, if the $t_r \in [0, T]$ are 1-separated, then $\sum_r |\sum_{n \leq N} a_n n^{-it_r}|^2 \ll (T + N) \log(2N) \sum |a_n|^2$.

Lemma 3.2 (Guth–Maynard [GM, Theorem 1.1]). Suppose $|b_n| \leq 1$ and $(t_r)_{r \leq R}$ are 1-separated points in $[0, T]$ with $|\sum_{n=N}^{2N} b_n n^{it_r}| \geq V$ for all $r \leq R$. Then $R \leq T^{o(1)} (N^2 V^{-2} + N^{18/5} V^{-4} + T N^{12/5} V^{-4})$.

Lemma 3.3 (Guth–Maynard [GM, Proposition 12.1], quoted verbatim). “Suppose $(b_n)_{n \sim N}$, $(t_r)_{r \leq R}$ are as in Theorem 1.1, and that $T^{5/6} \leq N \leq T$ and $V = N^\sigma$ with $\sigma \geq 7/10$. Then we have

$$R \lesssim N^{2-2\sigma} + T^{1/2} N^{3-4\sigma} + \inf_{k \in \mathbb{N}} \left(T^{\frac{k}{k+1}} N^{(4-6\sigma)\frac{k}{k+1}} + N^{(5-6\sigma)\frac{4k}{4k+3}} T^{\frac{2}{4k+3}} \right).$$

In particular, we have

$$R \lesssim N^{2-2\sigma} + T^{1/2} N^{3-4\sigma} + T^{(30\sigma-21)/5} N^{(46-60\sigma)/5}.$$

Here “Theorem 1.1” is [GM, Theorem 1.1], i.e. Lemma 3.2: the hypotheses are $|b_n| \leq 1$, the t_r are 1-separated in $[0, T]$, and $|\sum_{n=N}^{2N} b_n n^{it_r}| \geq V$ for every r . In [GM, §1.2], $A \lesssim B$ means that for every $\epsilon > 0$ there is $C(\epsilon)$ with $|A| \leq C(\epsilon) T^\epsilon B$ for all large T . The quotation was compared line by line with the arXiv text of [GM] (reports GA and R4; the closed form was also checked by R2 for the 0.548 version). We use the first display with $k \in \{1, \dots, 12\}$ only. Since the infimum is at most its value at any fixed k , each fixed k gives a valid bound, with a $\lesssim$ -constant independent of k (R4). The proof does not use the “in particular” display (the closed form used in the 0.548 version); it enters only one cross-check run (§4.7).

Lemma 3.4 (Ingham [In]; standard, see also [Ti]). $\int_0^T |\zeta(\frac{1}{2} + it)|^4 dt \ll T \log^4 T$.

Lemma 3.5 (Halász–Montgomery inequality; standard, see [Mo], [Iv]). If the $t_r \in [-T, T]$ are 1-separated and c_n is supported on $n \asymp N$, then $\sum_{r \leq R} |\sum_n c_n n^{-it_r}|^2 \ll (N + RT^{1/2}) \log(2T) \sum |c_n|^2$.

Lemma 3.6 (Approximation by partial sums; standard, see [Ti]). For $\sigma \geq \sigma_0 > 0$, $|t| < 2\pi y/C$ with $C > 1$ fixed: $\zeta(s) = \sum_{n \leq y} n^{-s} - \frac{y^{1-s}}{1-s} + O(y^{-\sigma})$.

Lemma 3.7 (Third-derivative test; standard, see [Ti], [GK]). Let f be real with $\lambda_3 \leq |f'''| \leq h\lambda_3$ on $[a, b]$. Then $\sum_{a < n \leq b} e(f(n)) \ll h^{1/2} (b-a) \lambda_3^{1/6} + (b-a)^{1/2} \lambda_3^{-1/6}$.

We also use two standard facts, see [Ti]: the truncated Perron formula and the convexity bound $\zeta(\sigma + it) \ll |t|^{(1-\sigma)/2+o(1)}$ for $\frac{1}{2} \leq \sigma \leq 1$.

Remark 3.8. We could not re-check theorem numbers in [Ti], [Mo], [Iv] and [GK] while writing. So the classical results are cited as standard, without numbers. The referee R2 confirmed that their statements are standard and correct. Lemmas 3.2 and 3.3 were checked against the arXiv text of [GM]: both by R2 and the earlier writer for the 0.548 version, and the full $\inf_k$ display again in phase II (GA, and this writer). The statements from [MT] were checked only through a summarising web tool (audit S2). An independent re-check (S3), also through such a tool, agrees.

4. THE REPLACEMENT FOR THE HEATH-BROWN-IWANIEC LEMMA

4.1. Statement. Let $T = x^\tau$. Let $A(s) = \sum \alpha_n n^{-s}$ with α_n supported in $(N_A, 2^{40}N_A]$ and $|\alpha_n| \leq d_K(n)(\log 2n)^K$. Define B in the same way with β_m and N_B . Let $C(s) = \sum_{N_C < n \leq 2N_C} c_n n^{-s}$ with $c_n \equiv 1$ or $c_n = \log n$. Write $N_A = x^a$, $N_B = x^b$, $N_C = x^c$ and $s_0 = a + b + c$. Constants may depend on K .

Lemma 4.1 (Replacement for [MT, Lemma 4.4]). *Let $\tau \in [0.45, 0.4565]$ and $s_0 \in [1 - 2 \cdot 10^{-4}, 1 + O(1/\log x)]$.*

(1) *For all $a, b \geq 0$ and $c > 0$: $\int_{T_0}^{\min(N_C, T)} |ABC(\frac{1}{2} + it)| dt \ll (N_A N_B N_C)^{1/2} (\log x)^{K_1}$.*

(2) *Suppose $c < \tau$, $a, b \in [0.087, 0.4565]$ and $c \geq 0.087$. Then*

$$\int_{N_C}^T |ABC(\frac{1}{2} + it)| dt \ll (N_A N_B N_C)^{1/2} (\log x)^{K_2}.$$

Moreover, the corner classes $\mathcal{Q}$ are those level classes (§4.4) with $|A| \geq N_A^{1/2} x^{-0.02}$ and $|B| \geq N_B^{1/2} x^{-0.02}$. All other classes together contribute $\ll (N_A N_B N_C)^{1/2} x^{-10^{-5}}$.

Compared with the 0.548 version, only the numbers change: there $\tau \leq 0.4521$, the lower length bound was 0.0958 and the saving off the corner was $x^{-10^{-4}}$. The rest of this section proves the lemma.

4.2. Small t and long C .

Lemma 4.2. *For $1 \leq t \leq N_C$, $|C(\frac{1}{2} + it)| \ll (\log x) N_C^{1/2}/t$.*

Proof. Apply Lemma 3.6 with $\sigma = \frac{1}{2}$ at $y = N_C$ and at $y = 2N_C$, and subtract. This gives $|C| \leq 2(2N_C)^{1/2}/|\frac{1}{2} - it| + O(N_C^{-1/2})$, and $N_C^{-1/2} \leq N_C^{1/2}/t$. For $c_n = \log n$, use partial summation on the same identity over $(N_C, y]$ with $y \leq 2N_C$. $\square$

Proof of Lemma 4.1(1). Let $U \in [T_0, \min(N_C, T)]$. The coefficients are divisor-bounded, so $\sum |\alpha_n|^2/n \ll (\log x)^{O(1)}$. Lemma 4.2, Cauchy-Schwarz and Lemma 3.1 give

$$\begin{aligned} \int_U^{2U} |ABC| &\ll (\log x)^{O(1)} \frac{N_C^{1/2}}{U} (U + N_A)^{1/2} (U + N_B)^{1/2} \\ &\ll (N_A N_B N_C)^{1/2} (\log x)^{O(1)} \left(\frac{1}{U} + \frac{1}{(UN_A)^{1/2}} + \frac{1}{(UN_B)^{1/2}} + \frac{1}{(N_A N_B)^{1/2}} \right). \end{aligned}$$

Each bracketed term is ≤ 1 . Summing over the $O(\log x)$ dyadic blocks proves (1). If $c \geq \tau$ the range in (2) is empty, so long C needs nothing more. $\square$

4.3. Pointwise caps and the transfer from ζ to C . The trivial bound gives $|A(\frac{1}{2} + it)| \ll N_A^{1/2} (\log x)^{O(1)}$, and likewise for B . For $t \geq 1$, apply Lemma 3.7 to $f(n) = -\frac{t}{2\pi} \log n$ on dyadic ranges, then partial summation:

$$(3) \quad |C(\frac{1}{2} + it)| \ll (\log x) (t^{1/6} + N_C^{1/2} t^{-1/6}).$$

Lemma 4.3 (Transfer lemma). *If the $t_r \in [N_C, T]$ are 1-separated, then $\sum_r |C(\frac{1}{2} + it_r)|^4 \ll T(\log x)^{12}$.*

Proof. Take $c_n \equiv 1$ first, and fix $t \geq N_C \geq 2$. Set $y_1 = \lfloor N_C \rfloor + \frac{1}{2}$, $y_2 \in (\lfloor N_C \rfloor, \lfloor 2N_C \rfloor] + \frac{1}{2}$, $\kappa = \frac{1}{2} + 1/\log x$ and $U = 2t$. Apply the truncated Perron formula at y_1 and at y_2 and subtract. Since $y_j \in \mathbb{Z} + \frac{1}{2}$, the truncation error is $\ll N_C^{1/2} (\log x)/U \ll \log x$.

Now move the contour to $\operatorname{Re} w = 0$:

- The pole of $\zeta(\frac{1}{2} + it + w)$ at $w = \frac{1}{2} - it$ contributes $\ll N_C^{1/2}/t \leq 1$.
- At $w = 0$, $(y_2^w - y_1^w)/w$ is regular.
- By convexity the horizontal segments are $\ll (3t)^{1/4+o(1)} N_C^{1/2}/U \ll 1$, using $t \geq N_C$.

On $\operatorname{Re} w = 0$ we have $|y_2^{iv} - y_1^{iv}|/|v| \leq k(v) := \min(\log 3, 2/|v|)$. Hence

$$\left| \sum_{N_C < n < y_2} n^{-1/2-it} \right| \leq \frac{1}{2\pi} \int_{-2t}^{2t} |\zeta(\tfrac{1}{2} + i(t+v))| k(v) dv + O(\log x).$$

The weight $\log n$ costs a factor $\ll \log x$, by partial summation in y_2 . By Hölder, using $\int_{-2t}^{2t} k \ll \log x$,

$$|C(\tfrac{1}{2} + it)|^4 \ll (\log x)^{4+3} \int_{-2t}^{2t} |\zeta(\tfrac{1}{2} + i(t+v))|^4 k(v) dv + (\log x)^8.$$

For 1-separated t_r , $\sum_r k(w - t_r) \ll \log x$ uniformly in w . Summing over r and using Lemma 3.4 on $[-T, 3T]$ gives $\sum_r |C(\tfrac{1}{2} + it_r)|^4 \ll (\log x)^8 T \log^4 T + R(\log x)^8 \ll T(\log x)^{12}$. $\square$

The value of τ enters this proof only through $T \leq x$; so Lemma 4.3, like (3) and Lemma 4.1(1), holds verbatim in the larger range of τ used here.

4.4. Discretisation and level classes. Cover $[N_C, T]$ by the unit intervals $I_j = [N_C + j, N_C + j + 1]$, $j \geq 0$, starting at N_C . Let t_j maximise $|ABC(\tfrac{1}{2} + it)|$ on $I_j \cap [N_C, T]$, so every $t_j \in [N_C, T]$. Then $\int_{N_C}^T |ABC| \leq \sum_j |ABC(\tfrac{1}{2} + it_j)|$. The points with j even, and those with j odd, are each 1-separated.

Sort the points into dyadic classes $|A(\tfrac{1}{2} + it_j)| \in (\tfrac{1}{2}x^{\lambda_A}, x^{\lambda_A}]$, and likewise λ_B for B and γ for C . There are $O((\log x)^3)$ classes with all levels $\geq -\tfrac{1}{2}$, plus three low classes, in which $|A|$, $|B|$ or $|C|$ is $\leq x^{-1/2}$. Write $R = x^\rho$ for the number of points in a class and set

$$E := \rho + \lambda_A + \lambda_B + \gamma - s_0/2.$$

A class contributes $\leq x^{s_0/2+E}$, so it suffices to show $E \leq -\eta + o(1)$ for every class.

Clamping. A class level may exceed its cap ($a/2$, $b/2$, or (3) for C) by $o(1)$, and s_0 may exceed 1 by $O(1/\log x)$. We clamp each class's lower level to its cap, and c to the end of its range. The count bounds stay valid, since they are non-increasing in λ . The exponent E changes by $o(1)$, which the fixed margin absorbs.

Low classes. By the caps, $\lambda_A \leq a/2 + o(1)$, $\lambda_B \leq b/2 + o(1)$ and $\gamma \leq c/2 + o(1)$. If, say, $\lambda_A < -\tfrac{1}{2}$, then $\rho \leq \tau + o(1)$ gives $E \leq \tau - \tfrac{1}{2} - \tfrac{a}{2} + o(1) < 0$.

Normalisation. Let

$$\mathcal{X} = \{A^k, B^k \ (k \leq 3), \ (AB)^k \ (k \leq 2), \ C^k \ (k \leq 6), \ AC, \ BC\}.$$

Each $X \in \mathcal{X}$ has length x^{n_X} . Its coefficients are $\ll d_{O(1)}(n)(\log x)^{O(1)} = x^{o(1)}$, and it is supported in $O(1)$ dyadic ranges. On a class, $|X| \geq x^{\lambda_X - o(1)}$, where λ_X is the corresponding sum of levels (e.g. $\lambda_{AB} = \lambda_A + \lambda_B$).

Pigeonhole a positive proportion of the points onto one dyadic piece $\sum_{n \sim N} c_n n^{-1/2-it}$. Rescale to $b_n = c_n(N/n)^{1/2}/\max |c_m|$, so that $|b_n| \leq 1$. Lemmas 3.2–3.3 then apply with $V = N^\sigma$ and $\sigma = \tfrac{1}{2} + \lambda_X/n_X - o(1)$.

Box edges. Lemma 3.2 has no range hypotheses. Lemma 3.3 needs $\sigma \geq 7/10$ and $T^{5/6} \leq N \leq T$. The certificate uses it, in the form E3k of §4.5, only on boxes where these hold at the nominal exponents, with a strict slack in σ .

- *The σ -edge.* E3k is used only on boxes with $\lambda_0 \geq n_1/5 + 10^{-6}$ (exact test). On such a box $\lambda \geq \lambda_0$, $n \leq n_1$ and $n_1 < 1$, so $\lambda/n \geq 1/5 + 10^{-6}/n_1 > 1/5 + 10^{-6}$. Hence $\sigma = \tfrac{1}{2} + \lambda_X/n_X - o(1) \geq 7/10 + 10^{-6} - o(1) > 7/10$ for x large, uniformly, since the boxes are finitely many and fixed. No edge argument is needed. In fact the slack used is much larger: on every box where the certificate uses E3k, $\lambda_0 - n_1/5 \geq 1571/2560000 \approx 6.1 \cdot 10^{-4}$ (measured by R4).
- *Why a slack.* The 0.548 version handled $\sigma \in [7/10 - o(1), 7/10)$ by switching to Lemma 3.2. That works for the closed form of Lemma 3.3 but not for the $\inf_k$ display, which can be smaller than Lemma 3.2 at $\sigma = 7/10$: at $N = T$, Lemma 3.2 gives $N^{4/5}$ while the $k = 1$ term gives $N^{0.6} + N^{0.7} + N^{0.4} + N^{5.2/7}$ (found by GA, confirmed by R4). So that step is not used here.

- *The lower N -edge* is automatic. E3k is used only where $n_0 \geq 5\tau_{\text{hi}}/6$, so every dyadic piece has $N \geq x^{n_0} \geq T^{5/6}$.
- *The upper N -edge*. If $N \in (T, 2^{O(1)}T]$, apply Lemma 3.3 with $T' = N$. Every power of T in E3k has exponent in $[0, 1]$, so the bound changes by a factor $O(1)$.

In each case the exponents move by $o(1)$, which the margin absorbs.

4.5. Count bounds and monotonicity. Fix $X \in \mathcal{X}$ with length exponent n and level $\lambda > 0$, and put $\sigma = \frac{1}{2} + \lambda/n$. Up to $+o(1)$, the following hold.

$$(E1, \text{Lemma 3.1}) \quad \rho \leq \max(\tau, n) - 2\lambda;$$

$$(E2, \text{Lemma 3.2}) \quad \rho \leq \max\left(n - 2\lambda, \frac{8}{5}n - 4\lambda, \tau + \frac{2}{5}n - 4\lambda\right);$$

$$(E3k, \text{Lemma 3.3}, k \geq 1) \quad \rho \leq \max\left(n - 2\lambda, \frac{\tau}{2} + n - 4\lambda, \frac{k}{k+1}(\tau + n - 6\lambda), \frac{4k}{4k+3}(2n - 6\lambda) + \frac{2\tau}{4k+3}\right) \\ \text{if } \frac{5}{6}\tau \leq n \leq \tau \text{ and } \lambda \geq n/5 + 10^{-6}, \text{ i.e. } \sigma \geq \frac{7}{10} + 10^{-6}/n;$$

$$(E4, \text{Lemma 4.3}, X = C) \quad \rho \leq \tau - 4\gamma; \quad \text{and trivially } \rho \leq \tau.$$

E3k is the first display of Lemma 3.3 at a fixed k , written in exponents ($n = \log_x N$, $\tau = \log_x T$): since $\sigma n = n/2 + \lambda$, we have $(2 - 2\sigma)n = n - 2\lambda$, $(3 - 4\sigma)n = n - 4\lambda$, $(4 - 6\sigma)n = n - 6\lambda$ and $(5 - 6\sigma)n = 2n - 6\lambda$. The certificate uses $k = 1, \dots, 12$. The slack 10^{-6} in the hypothesis is the one explained under *Box edges* (§4.4). The closed form of Lemma 3.3 is not among these estimates.

Monotonicity. Every piece of E1, E2, E3k and E4 is non-decreasing in n and in τ and non-increasing in λ . For E3k this is immediate, since each piece is linear: the coefficients of n are $1, 1, \frac{k}{k+1}, \frac{8k}{4k+3}$, those of τ are $0, \frac{1}{2}, \frac{k}{k+1}, \frac{2}{4k+3}$, all ≥ 0 , and those of λ are negative. So on a box $n \in [n_0, n_1]$, $\lambda \geq \lambda_0$, $\tau \leq \tau_{\text{hi}}$, each estimate is at most its value at $(n_1, \lambda_0, \tau_{\text{hi}})$. Also $\sup_{\text{box}} \min_e \leq \min_e \sup_{\text{box}}$; in fact the certificate accepts a box on a single estimate, so no interchange of sup and min is needed (R4).

4.6. The corner classes. Put $\delta_A = a/2 - \lambda_A$ and $\delta_B = b/2 - \lambda_B$. The corner classes are those with $\delta_A, \delta_B \leq \kappa := 0.02$.

Case $\min(a, b) \geq 0.276$. Apply Lemma 3.5 to A with $c_n = \alpha_n n^{-1/2}$, so that $G := \sum |c_n|^2 \ll (\log x)^{O(1)}$, and take $V = \frac{1}{2}N_A^{1/2}x^{-\delta_A}$.

- $V^2 \geq x^{a-2\kappa}/4 \geq x^{0.236}/4$, which exceeds a large multiple of $GT^{1/2} \log T \leq x^{0.22825+o(1)}$.
- So $R \ll GN_A V^{-2} \log T \ll (\log x)^{O(1)} x^{2\delta_A}$. The same argument for B gives $R \ll (\log x)^{O(1)} x^{2\delta_B}$.
- Since $|C| \ll N_C^{1/2} \log x$, the class contributes $\ll (\log x)^{O(1)} x^{s_0/2 + \min(2\delta_A, 2\delta_B) - \delta_A - \delta_B} \leq (\log x)^{O(1)} x^{s_0/2}$.

This is a loss of a power of $\log x$ only, with no power saving. In the application it is absorbed by Lemma 2.3. This case works whenever $0.236 > \tau/2$, i.e. for $\tau < 0.472$.

Case $\min(a, b) < 0.276$.

- Here $c > 1 - 2 \cdot 10^{-4} - 0.276 - 0.4565 = 0.2673$.
- By (3), $c/2 - \gamma \geq \min(c/2 - \tau/6, c/6) - o(1) \geq \min(0.0575, 0.04455) - o(1) = 0.04455 - o(1)$.
- Lemma 3.1 applied to AB , of length $a + b > 1 - 2 \cdot 10^{-4} - \tau > \tau$, gives $\rho \leq 2\delta_A + 2\delta_B + o(1)$.
- Hence $E \leq \delta_A + \delta_B - 0.04455 + o(1) \leq 2\kappa - 0.04455 + o(1) = -0.00455 + o(1)$.

This case works whenever $(0.724 - 2 \cdot 10^{-4} - \tau)/6 > 2\kappa$ and $(0.724 - 2 \cdot 10^{-4} - \tau)/2 - \tau/6 > 2\kappa$, i.e. for $\tau < 0.483$.

4.7. The computer certificate. All remaining classes are handled by the script `numerics/certify_exact.k.py`¹. It is a branch-and-bound over boxes in $(a, b, \lambda_A, \lambda_B, \gamma)$, in exact rational arithmetic. It is the exact certifier `numerics/certify_exact.py` of report X1 (MD5

¹MD5 8c4aded7ba48509ceafb3480a2be9bf3. Invocation:

TLO=0.45 THI=0.4565 MARGIN=0.00001 EPSP=0.0002 VSLACK=0.000001 MAXIT=150

python3 certify_exact.k.py min0k.noclosed 0.02: certified after 67 rounds (iterations 0–66), 331 054 leaves; log numerics/run_min0knc.4565.out.

f083c952... , unchanged and imported), with the estimates E3k added. According to R4’s diff, nothing else changed: the tiling, the empty and corner tests, the caps, the tails and the acceptance test are X1’s. One documentation slip: line 8 of the script’s docstring describes the E3k validity test as $\lambda_0 \geq n_1/5$, as for the closed form. The code also requires the slack, $\lambda_0 \geq n_1/5 + \text{VSLACK}$ with $\text{VSLACK} = 10^{-6}$, which is the test stated here (R4, issue m2; the code is correct).

Setup.

- On a box, c ranges over $[1 - a_1 - b_1 - 2 \cdot 10^{-4}, 1 - a_0 - b_0]$, which covers $s_0 \in [1 - 2 \cdot 10^{-4}, 1]$.
- $\tau \in [\tau_{\text{lo}}, \tau_{\text{hi}}] = [9/20, 913/2000]$ is treated uniformly, and every bound uses τ_{hi} . The range conditions of E3k are checked as $n_1 \leq \tau_{\text{lo}}$, $n_0 \geq 5\tau_{\text{hi}}/6$ and $\lambda_0 \geq n_1/5 + 10^{-6}$ (§4.4).
- The initial boxes are (a, b) -cells of width $1/50$ starting at $1 - 2\tau_{\text{hi}} = 0.087$, with $a \leq b$ by symmetry, together with $\lambda_A \in [-\frac{1}{2}, a_1/2]$, $\lambda_B \in [-\frac{1}{2}, b_1/2]$ and $\gamma \in [-\frac{1}{2}, \frac{1}{2}]$.

The test on each box.

- $\bar{\rho}$ is the minimum, over $X \in \mathcal{X}$ and over E1, E2, E3k ($k \leq 12$), E4 and the trivial bound, of the box suprema from §4.5. These are evaluated at the lower level ends. A polynomial whose level lower end is ≤ 0 is not used.
- The contribution exponent uses the upper ends of the level ranges. These are capped by $a_1/2$ and $b_1/2$, and for C by

$$\min(c_1/2, \max(\tau_{\text{hi}}/6, c_1/3)).$$

- The box is certified if $E_{\text{box}} := \max(\bar{\rho}, 0) + \bar{u}_A + \bar{u}_B + \bar{u}_C - (1 - 2 \cdot 10^{-4})/2 \leq -10^{-5}$.

Exempt boxes. Two kinds of boxes are not tested:

- empty boxes, where a level exceeds its cap or $a, b < \tau_{\text{hi}}$, $0.087 \leq c < \tau$ fails;
- boxes inside $\mathcal{Q}$, which §4.6 covers.

The run. Floats only decide which boxes to try and in what order; every leaf is accepted only after an exact test. Uncertified boxes are bisected along their widest weighted side.

- The run ends with no uncertified box after 67 rounds (iterations 0–66). The leaves tile the domain exactly: 331 054 leaves, namely 807 empty, 3 310 in the corner region and 326 937 certified.
- The largest certified value is $E_{\text{box}} \leq -18999/1884160000 \approx -1.008 \cdot 10^{-5}$. It sits on the five-factor box, $(a, b) \approx (0.400, 0.400)$, $\gamma \approx 0.0434$.
- The low classes (some level $< -\frac{1}{2}$) are certified in closed form, with $E \leq -9779/60000 \approx -0.163$ and $E \leq -869/10000$.
- Among the $\inf_k$ terms, $k = 5$ and $k = 4$ win most often. This agrees with the five-factor analysis, which uses $k = 5$.
- *Independent re-verification* (R4). The certifier stores no leaves, so its soundness rests on re-running it. The referee R4 therefore wrote `rounds/round2/R4_scripts/r4_verify.py` (MD5 471a01049eeb3af168806a92bce61ec6). It records every accepted leaf of the `min0k_noclosed` run and re-checks it with independently written code that does not call the certifier’s evaluation functions:
 - the bounds are written in σ -form, directly from the cited statements;
 - box suprema are taken by enumerating the vertices $\{n_0, n_1\} \times \{\lambda_0, \lambda_1\} \times \{\tau_{\text{lo}}, \tau_{\text{hi}}\}$. This is valid because each candidate is a maximum of affine functions, and it does not rely on the monotonicity claim of §4.5;
 - the E3k hypotheses are checked with strict $\sigma > 7/10$;
 - the empty and corner tests, the caps and the acceptance test are recomputed;
 - the exact total volume of the leaves is checked to equal that of the initial cells, so no box is lost.
 Result: 331 054 leaves at $\tau_{\text{hi}} = 913/2000$ and 586 580 at $\tau_{\text{hi}} = 114173/250000 = 0.456692$, with 0 failures and with equal volumes in both runs. The maxima agree with the certifier’s. Usage: run it with `PYTHONPATH=numerics` from a scratch directory, since it writes a JSON file.
- *Re-runs.* R4 re-ran the certifier for `min0k_noclosed` and `min0k` at $913/2000$ and for `min0k_noclosed` at 0.456692 , and obtained bit-identical leaf counts. The writer of this note also re-ran the headline

invocation from fresh copies of the three scripts (same MD5 sums), in a separate directory. It reproduced the same leaf counts and the same maximum.

- Rigour of the box bounds: they follow from the monotonicity in §4.5. Since there are finitely many boxes, the $o(1)$ terms are absorbed by the fixed margin 10^{-5} . Levels are clamped to their caps as in §4.4.
- Scope: the run checks only the arithmetic of the box bounds, given E1–E4, the caps and the corner lemma of §4.6.

Inputs actually used. The run is called `min0k.noclosed`. Its inputs are MVT, Lemma 3.2, the $\inf_k$ display of Lemma 3.3 ($k \leq 12$) and Lemma 4.3. The following are *not* used: the closed form of Lemma 3.3, Huxley’s estimate, Jutila’s estimate, the twelfth moment, Bourgain’s bound for ζ , and Vinogradov–Korobov for C .

Cross-checks and sanity runs (same code and settings). None of these is used in the proof.

- `min0k` adds the closed form of Lemma 3.3, with its old test $\lambda_0 \geq n_1/5$; for the closed form alone, the σ -edge step of the 0.548 version is valid. Result: certified, 330 937 leaves. Adding the full input set of the earlier code as well (`fullk`): certified, 310 004 leaves. Both take 67 rounds.
- Regression: with the old input set `min0` at $\tau_{hi} = 0.4521$, the new code reproduces the tiling of the 0.548 version exactly (98 433 leaves: 661 empty, 2 432 corner, 95 340 certified).
- In every run, the float mirror of the new code agrees with the float code of the 0.548 version on all old estimates, with maximum difference 0.0. That code is `numerics/certify_548.py` (MD5 4d0afc03...).
- The sanity run of the 0.548 version failed at $\vartheta = 0.5435$, with $\max E = +0.0019$ at $(a, b, c) = (0.4, 0.4, 0.2)$. That run used the closed form of Lemma 3.3. With the closed form, the five-factor box closes only for $\vartheta > 0.5437644$ (§6). The present certificate, with the $\inf_k$ display, supersedes it.
- The control run at $\vartheta = 0.5433 < 69/127$ fails (Remark 5.1).
- In the 0.548 version, a run without Lemma 4.3 failed at $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ by $+0.025$. So the transfer of the fourth moment from ζ to C is essential.

This completes the proof of Lemma 4.1: the corner classes are handled in §4.6 and all others by the certificate. $\square$

5. PROOF OF THEOREM 1.1

Parameters. Let $\vartheta > 0.5435$ and $\varepsilon_0 > 0$. We may assume $\vartheta \leq 0.55$, since for $\vartheta > 0.55$ the claim is [MT, Theorem 1.1].

- First fix $\varepsilon := \min(\varepsilon_0, (\vartheta - 0.5435)/4, 10^{-4})$ and $\theta := \vartheta - \varepsilon$. Then $\theta \in [0.5435 + 3\varepsilon, 0.55 - \varepsilon]$. These depend only on ϑ and ε_0 , not on H . (The 0.548 version divided by 2; dividing by 4 keeps b strictly below 0.4565, see below.)
- Then, as in [MT, §4.1], split the sum into short sums. This reduces the theorem to intervals $(y, y+H']$ with $y \geq x$ and $H' \in [y^{\theta+\varepsilon}, 2y^{\theta+\varepsilon}] \subset [y^\varepsilon, y]$, where Lemma 2.1 applies. This holds also when $H > x$. We write x, H for y, H' .

Run MT’s argument (§2) with these θ, ε . By Lemma 2.1 the saving is $(\log x)^{-1/3+\varepsilon}$, and $\varepsilon \leq \varepsilon_0$. By Lemma 2.2 it suffices to prove (2) for each configuration (1). The upper limit of integration is $T = x^{1+\varepsilon/10}/H \leq x^\tau$ with $\tau := 1 - \theta - 0.9\varepsilon = 1 - \vartheta + 0.1\varepsilon \in [0.45, 0.4565]$.

Removing P and R . Factors with $N_i < 2$ have at most three terms and are $O(1)$; drop them. Write $N_i = x^{\alpha_i}$ and $s_0 = \sum \alpha_i$. Then $s_0 \in [1 - \varepsilon/2 - o(1), 1 + O(1/\log x)]$, because $R \leq x^{\varepsilon/2}$ and $P_1 = x^{o(1)}$. In particular $s_0 \geq 1 - 2 \cdot 10^{-4}$. We have:

- $|P(\frac{1}{2} + it)| \ll_A P_1^{1/2}(\log x)^{-A}$ on $[T_0, T]$, by Lemma 2.3;
- $|R(\frac{1}{2} + it)| \ll R^{1/2}(\log x)^{O(1)}$, trivially;
- $P_1 R x^{s_0} \asymp x$.

So it suffices to show $\int_{T_0}^T |N_1 \cdots N_{2k-1}(\frac{1}{2} + it)| dt \ll x^{s_0/2}(\log x)^{O(1)}$.

Type II. Let $w_1 = 1 - \theta$ and $w_2 = \theta - \varepsilon$. Suppose some subsum $\sum_I \alpha_i \in [w_1, w_2]$. Then $X_I \geq x^{1-\theta} \geq T$ and $X_{I^c} \geq x^{s_0-\theta+\varepsilon} \geq x^{1-\theta} \geq T$, and Lemma 2.4 applies.

Otherwise. Apply Lemma 2.5 (note $s_0 > w_2$). This gives I , r and J with

$$a := \sum_I \alpha_i < 1 - \theta = 1 - \vartheta + \varepsilon < 0.4565, \quad c := \alpha_r > 2\theta - 1 - \varepsilon \geq 0.087 + 5\varepsilon > 0.087,$$

$$b := \sum_J \alpha_j < s_0 - w_2 \leq 1 - \vartheta + 2\varepsilon + O(1/\log x) \leq 0.4565 - \frac{1}{2}(\vartheta - 0.5435) + O(1/\log x).$$

- Since $c > 1/20 + o(1)$ and the Möbius-type factors have length $\leq (2x)^{1/20}$, the factor N_r is smooth: $r \leq k - 1$.
- Put $A = \prod_I N_i$, $B = \prod_J N_j$ and $C = N_r$. These satisfy the hypotheses of §4 with $K = 40$.
- Lemma 4.1(1) covers $[T_0, \min(N_C, T)]$.
- If $c < \tau$, Lemma 4.1(2) covers $[N_C, T]$. Its hypotheses hold:
 - $a > w_2 - c > \theta - \varepsilon - \tau = 2\theta - 1 - 0.1\varepsilon \geq 0.087 + 5.9\varepsilon > 0.087$;
 - $b > s_0 - w_1 - \tau \geq 2\theta - 1 + 0.4\varepsilon - o(1) \geq 0.087$;
 - $a < 0.4565$; $b < 0.4565$ for x large, by the display above; and $0.087 \leq c < \tau$.

In every case the integral is $\ll x^{s_0/2}(\log x)^{O(1)}$. Combined with the factor $(\log x)^{-A}$ from P , this gives (2), and Theorem 1.1 follows. We never use Lemma 4.4 of [MT]. $\square$

Remark 5.1 (Closer to 69/127). The same proof gives Theorem 1.1 for somewhat smaller ϑ , at the price of smaller margins. We keep the headline at 0.5435, where the margin is 10^{-5} . Each item below uses the same code, input set `min0k_noclosed` (also certified with `min0k`), $\kappa = 1/50$, and the corner lemma, which holds for $\tau < 0.472$.

- $\vartheta > 0.5434$: $\tau_{\text{hi}} = 2283/5000$, $\text{EPSP} = 10^{-4}$, margin 10^{-5} and $\varepsilon := \min(\varepsilon_0, (\vartheta - 0.5434)/4, 10^{-4})$. This EPSP is valid, since $s_0 \geq 1 - \varepsilon/2 - o(1) > 1 - 10^{-4}$. The lower length bound is 0.0868. Certified after 76 rounds with 388 691 leaves (388 580 for `min0k`). Corner case (ii): $c > 0.2673$.
- $\vartheta > 0.543308 = 69/127 + 9.1 \cdot 10^{-7}$: $\tau_{\text{hi}} = 114173/250000 = 0.456692$, $\text{EPSP} = 2 \cdot 10^{-7}$, margin $3 \cdot 10^{-7}$, σ -slack 10^{-7} , and $\varepsilon := \min(\varepsilon_0, (\vartheta - 0.543308)/4, 10^{-7})$, so that $s_0 > 1 - 2 \cdot 10^{-7}$. The lower length bound is 0.086616. Certified after 103 rounds with 586 580 leaves (828 empty, 3 207 corner, 582 545 certified); 586 479 for `min0k`. The invocation is that of the footnote in §4.7 with `THI=0.456692`, `MARGIN=0.0000003`, `EPSP=0.0000002` and `VSLACK=0.0000001`. The largest certified value is $E_{\text{box}} \leq -4565757/15073280000000 \approx -3.03 \cdot 10^{-7}$, and the slack used for E3k is at least $\lambda_0 - n_1/5 \geq 1.38 \cdot 10^{-3}$ (R4). Corner lemma: $0.236 > \tau/2 = 0.228346$; in case (ii), $c > 0.267308$ and $c/6 > 0.04455 > 2\kappa$.
- Runs at $\tau_{\text{hi}} = 0.45665$, 0.45668 and 0.45669 ($\text{EPSP } 2 \cdot 10^{-5}, 4 \cdot 10^{-6}, 10^{-6}$) are also certified.
- *Control.* At $\tau_{\text{hi}} = 0.4567$, i.e. $\vartheta = 0.5433 < 69/127$, with EPSP, margin and slack all 10^{-7} , the run is not certified. The number of open boxes doubled every round, and the run was stopped after round 84 with 317 127 open boxes and float maximum $E = +10^{-5}$. An exact pointwise evaluation at $(a, b, c) = (0.4, 0.4, 0.2)$, $\sigma_A = \sigma_B = 0.716525$, $\gamma = 0.043303$ and $\tau = 0.4567$ gives $E = +3 \cdot 10^{-6} > 0$. So the model itself fails there, and the certificate is not vacuous.
- 69/127 itself cannot be reached this way. At the binding point the slack is about $58/127 - \tau_{\text{hi}}$, so the margins would have to tend to 0.

These extensions have the same status as Theorem 1.1. The referee R4 checked them as well: the corner numbers and the $\varepsilon \leq 10^{-7}$ bookkeeping at 0.456692, a re-run, and the leaf-by-leaf re-verification of §4.7. R4 recommends keeping them as a remark rather than in the theorem.

6. LIMITS OF THE METHOD

The headline exponent 0.5435 lies $1.9 \cdot 10^{-4}$ above $69/127 = 0.5433071\dots$. Remark 5.1 reduces the gap to $9.1 \cdot 10^{-7}$. Below 69/127 the method itself fails, for the reason explained next.

Five smooth factors. The configuration $N_1 = \dots = N_5 = x^{1/5}$ with smooth coefficients has no subsum in $[1 - \theta, \theta]$ for $\theta < 3/5$. So it must be handled as $A = N_1 N_2$, $B = N_3 N_4$, $C = N_5$, i.e. $(a, b, c) = (0.4, 0.4, 0.2)$. The maximum of the certificate and the failing control run both sit at this

point. Internal report P2 treats this configuration; an internal referee (R1) judged it correct with fixable issues. Report PB1 (phase II) recomputed the binding point in exact rational arithmetic.

- The inputs are: MVT, Lemma 3.6, the cap (3), the fourth-moment bound for the fifth factor, and the $\inf_k$ form of Lemma 3.3 with $k = 5$. With these inputs the five-factor case closes exactly when $\vartheta > 69/127 = 0.5433071\dots$, i.e. $\tau < 58/127$.
- At $\vartheta = 69/127$ the pair products $N_i N_j$ have length $x^{2/5} \approx T^{0.88}$, the binding level is $\sigma = 91/127$, each $|N_i|$ has size $x^{11/254}$, and the number of points must stay below $x^{36/127}$. Two bounds equal $36/127$ exactly: MVT for a pair, equivalently Lemma 4.3 for N_5 ; and the part $N^{(5-6\sigma)20/23} T^{2/23}$ of the $k = 5$ term. The term $T^{1/2} N^{3-4\sigma}$ gives $179/635$, only $1/635 \approx 0.0016$ lower. (The writer re-derived these values by hand from E1, E3k and E4.)
- With the closed form of Lemma 3.3 instead, the threshold is $\vartheta > 0.5437644$, where $\tau = 1 - \vartheta$ is the smaller root of $30\tau^2 - 32.8\tau + 8.72 = 0$. This is why the 0.548 version, which used the closed form, failed at 0.5435 in its sanity run.

Independent certification. Internal report P4 used code written independently of P3's. It certified the generic level-set model, with A, B arbitrary and C smooth. Its inputs were MVT, Huxley, Lemma 3.2, the $\inf_k$ form of Lemma 3.3 ($k \leq 8$), Jutila ($k = 3$), the fourth and twelfth [HB] moments of ζ , and Bourgain's bound [Bo]. The results are in Table 1.

- $\vartheta = 0.5432$ was not certified within the box budget. This agrees with the exact certificate of §4.7, which passes at 0.5434 and fails at 0.5433.
- With pre-Guth–Maynard inputs only (MVT, Huxley, Jutila), the same code certifies 0.552 and fails at 0.548. This is consistent with the exponent 0.55 of [MT].
- P4 certifies only the level-set model. Its reduction to the top t -scale and its corner argument, via Huxley in Halász form, were argued by hand and classed as routine.

ϑ	0.548	0.545	0.544	0.5438	0.5435	0.5434
certified margin η	10^{-3}	10^{-3}	$3 \cdot 10^{-4}$	$3 \cdot 10^{-4}$	10^{-4}	$3 \cdot 10^{-5}$
boxes	12 164	30 298	33 477	39 149	43 639	46 242

TABLE 1. Independent certification of the level-set model (report P4, `numerics/p4_check.py`).

What blocks 69/127: findings of phase II. The statements below are model computations and structural observations from internal reports (PB1, PB2, PC1, SC5, DR4, S4). Unless marked, they were not re-derived by the writer, and they are not theorems.

- (a) *Reduction to one zeta segment* (PB1, and independently the control agent PC1, which had no access to the project's idea registry). Take all five smooth factors in the same dyadic block. This is a legitimate configuration of (1). Then $N_1 = \dots = N_5 = S_N$, a single segment $S_N(s) = \sum_{n \sim N} n^{-s}$ with $N \asymp x^{1/5}$, and $T = N^{5(1-\vartheta)}$. What is needed is

$$\int_{T_0}^T |S_N(\tfrac{1}{2} + it)|^5 dt \ll N^{5/2} (\log N)^B.$$

So below 69/127 the problem contains a fifth-moment problem for one segment of ζ . In large-values language (notation of the ANTEDB [TTY]), PB1 reports that going below 69/127 needs $\text{LV}_\zeta(\sigma, 2.3) < 5 - 5\sigma$ for $\sigma \in (0.70, 0.72]$. The record values it lists are 1.48, 1.46 and 1.435 at $\sigma = 0.705, 0.71$ and 0.715 , against targets 1.475, 1.45 and 1.425. So one would need to beat the record by 0.005–0.010. This is the obstacle O4sym of the project database.

- (b) *The joint route is refuted* (PB1, SC5). The 0.548 version suggested a first-moment bound $\sum_{t \in S_A(\sigma)} |C(\tfrac{1}{2} + it)| \lesssim |S_A(\sigma)|$ for C on the large-value set $S_A(\sigma)$ of A ; this would close the five-factor family for every $\vartheta > 1/2$. In the equal-segment configuration, however, $A = N_1 N_2 = C^2$. So A is large exactly where C is large, and the bound would force a pointwise bound of about

$T^{0.065}$ for a zeta segment, whereas the best known is about $T^{0.155}$ (cf. [Bo]). This also explains why an experiment at $x = 10^{10}$ found $|C|$ systematically larger on $S_A(\sigma)$. The observation came from the project’s set E9 (joint large values of A and C).

- (c) *Additive energy inside the Guth–Maynard method* (PB2). In the proof of [GM, Proposition 12.1], the k -dependent terms arise from one sum (called S_2 in report PB2). Even if an additive-energy input removed these terms entirely, the model frontier would be $19/35 = 0.542857$. It is a quadruple point at $\sigma = 5/7$, where MVT, the generic term $N^{5-6\sigma}$, the term $T^{1/2}N^{3-4\sigma}$ and the requirement meet; $19/35$ solves $5\vartheta - 2 = 1 - 5(1 - \vartheta)/8$. So this route gains at most $69/127 - 19/35 \approx 0.00045$. Below $19/35$ no value of the energy helps, and the energy is not binding at $69/127$.
- (d) *Range of σ* (obstacle O2a, from the 0.548 version). Below ≈ 0.5433 the relevant range of σ extends below $7/10$, outside Proposition 12.1. In a sampled version of the model, Montgomery’s large values conjecture on $\sigma \in [0.70, 0.73]$ alone gains only about 0.004. The full large values conjecture would push the model to $\vartheta \rightarrow 1/2$, apart from the corners.
- (e) *Priced targets* (entry N7 of the project database; reports PB2, DR4, S4 and PC1). Table 2 lists what some hypothetical inputs would give in the model. These are prices, not results.

hypothetical input (exponent model)	effect on the threshold
none (this note)	$69/127 = 0.54331$
k -dependent terms of GM’s S_2 removed (best case of an energy input)	$19/35 = 0.54286$
a large values bound at the binding level beating the fourth-moment level by $T^{-0.002}$	0.5424
the same with $T^{-0.005}$	0.5410
the same with $T^{-0.01}$	0.5389
MVT for pairs of segments improved by $T^{-0.0117}$	0.5382
sixth moment of ζ on $\{ \zeta \geq T^\beta\}$ bounded by $T^{1+o(1)}$ down to $\beta = 0.09$	gain 0.0033
full sixth moment $\int_0^T \zeta(\frac{1}{2} + it) ^6 dt \ll T^{1+o(1)}$	gain 0.0165

TABLE 2. Model prices for going below $69/127$ (internal reports; not theorems). For the restricted sixth moment, the bound is known with $\beta = 11/72$ [TTY].

6.1. Below $69/127$: a refereed reduction. The following result (SUB542) comes from report GA2. It was refereed by R5 with verdict *accept with corrections* (one definitional correction, M1, and two minor ones, m2 and m3). We state it in R5’s corrected form. It is a reduction, not an improvement of Theorem 1.1: the residual family contains the five-factor configuration that blocks $69/127$.

Proposition 6.1 (SUB542). *Let $\theta = 0.542$, $\tau_{hi} = 0.458$ and $\text{LOW} = 1 - 2\tau_{hi} = 0.084$, and put $\text{NE}[R] := \{(a, b) : \max(|a - 0.4|, |b - 0.4|) \leq R\}$.*

- (i) *The conclusion of Lemma 4.1(2), with $\tau \in [0.45, 0.458]$, $s_0 \geq 1 - 10^{-4}$, $a, b \in (0.084, 0.458)$ and $0.084 < c < \tau$, holds for every triple with $(a, b) \notin \text{NE}[0.0075]$. The proof is an exact certificate: the code of §4.7, changed only so that boxes contained in the closed box $\text{NE}[0.0075]$ are set aside as “excluded” and never counted as certified (`numerics/certify_exact_k_sub.py`, MD5 00767d5a...; EPSP = 10^{-4} , margin 10^{-5} , slack 10^{-6} , $\kappa = 1/50$). With `min0k` it uses 425 506 leaves (698 empty, 2 835 corner, 418 542 certified, 3 431 excluded) in 65 rounds, with largest certified value $-13/1280000$. Without the closed form (`min0k_noclosed`) it uses 425 560 leaves. R5 re-ran both runs; the first is bit-identical to GA2’s log.*
- (ii) *Consequently, for $\vartheta \in (0.542, 0.55]$ and $\varepsilon := \min(\varepsilon_0, (\vartheta - 0.542)/4, 10^{-4})$, the argument of §5 closes every configuration (1) except those in the residual set Res. A configuration is in Res if it has no subsum in $[1 - \theta, \theta - \varepsilon]$, every smooth factor has length $< \tau$, and every admissible split (a, b, c) satisfies $(a, b) \in \text{NE}[0.0075]$. Configurations with a smooth factor of length $\geq \tau$ are closed by Lemma 4.1(1); this is R5’s correction M1. For $\alpha \in \text{Res}$ the MT split of §5 exists and*

is admissible. Hence Theorem 1.1 of [MT] for every $\vartheta > 0.542$ reduces to Lemma 4.1(2) on the near-equal family $\text{NE}[0.0075]$.

The residual family (R5). A configuration in Res has a smooth factor C of length $c \in [0.185 - \varepsilon/2 - o(1), 0.215 + o(1)]$ and two groups A, B of total lengths 0.4 ± 0.0075 . No factor in A or B has length in $(0.015, \theta - \varepsilon - b]$, an interval which contains $(0.015, 0.1345]$. For exactly five smooth factors of lengths $0.2 + \delta_i$, the configuration is open if and only if $|\delta_i + \delta_j| \leq 0.0075$ for all $i \neq j$. This forces $|\delta_i| \leq 4R/3 + O(\varepsilon) = 0.01$. R5 proved this criterion in general and checked it by brute force on 40 000 random configurations, with no mismatch.

GA2 also certified the analogous statements with $R = 0.0125$ at $\theta = 0.541$ (655 573 leaves) and with $R = 0.02$ at $\theta = 0.540$ (532 758 leaves). These two runs were not re-run by the referee.

What the remaining case needs. The following are model-level analyses from internal reports (entries O4sym, PC3-EQ, PB4-F and BB-MAP of the project database), not theorems.

- The residual case is the equal-segment problem of item (a) above. The one live target is to beat the MVT (fourth-moment) count by a power $T^{-\delta}$ for products of two smooth segments of length $x^{1/5}$ at $\sigma \in [0.709, 0.717]$. A restricted moment of order about 4.03 (4.08 at $\theta = 0.540$) would suffice; the sixth moment is not needed.
- For uniform slices, the control agent PC3 finds this equivalent, up to T^ε , to improving Ingham's large values bound for ζ itself. That bound gives $\ll T^{1-4v+o(1)}$ points with $|\zeta(\frac{1}{2} + it)| \geq T^v$, and the relevant heights are T^v with $v \in [0.0909, 0.0952]$, i.e. $T^{0.09-0.095}$. No such improvement is known to us; PC3 states this from memory, unchecked.
- A multi-scale, Euler-product coherence statement would suffice (PB3-ITE). It says: if $|S_N(\frac{1}{2} + it)| \geq V = T^v$, then some segment $S_{N'}$ with $N' \leq T^{1/3}$ has $|S_{N'}(\frac{1}{2} + it')| \geq V^{1-\delta_I}$ at a point with $|t - t'| \leq 2$. With $k = 4$ and $\delta_I \leq 0.496$ it would give $\vartheta^* \approx 0.540$ (PB4).
- The statement is strongly supported by numerics: the median of $1 - \delta_I$ is 0.95–0.96 for $N = 100$ –316 and $T \leq 5.6 \cdot 10^5$, though only at the scale of typical fluctuations. It follows from the Lindelöf hypothesis, but no unconditional proof is known. The obvious route through the approximate functional equation is circular (PB3).

7. PROVENANCE AND STATUS

This note was produced by an automated multi-agent proof search run by Claude (Anthropic). It followed a strategy co-authored by Siddharth Iyer. It is the phase II version of a note that proved the same statement for $\vartheta > 0.548$.

Sources. The mathematics comes from internal reports, written and checked by separate agents. Phase I: the audit of [MT] (S2) and its re-check (S3); the five-factor lemma (P2, refereed by R1); the general Lemma 4.1 and the assembly (P3, refereed by R2); the exact-arithmetic certificate (X1); and the independent certification (P4). Phase II: report GA (the $\inf_k$ estimate, the certificate `certify_exact_k.py`, the box-edge correction, the corner numbers at the new τ , and the parameter choice); and the limit analyses PB1, PB2, PC1 (control), SC5, DR4 and S4. Later phase II rounds contributed the following:

- GA2, the sub-family certificate of §6.1, refereed by R5;
- PB3, PB4, PC3 (control) and BB, the analyses quoted at the end of §6.

The writer of this note (W2) collated these reports. The writer also re-derived by hand: the exponent form of the $\inf_k$ display and its monotonicity; the σ -slack of the box edges and the counterexample to the old edge argument; the corner numbers; the parameter inequalities of §5; and the values at the five-factor binding point. The writer re-ran the headline certificate and obtained the same result. Statements that the reports do not support were omitted or are marked above.

Refereeing. The 0.548 version: an independent agent (R2), who did not write the proof, refereed the proofs of Lemma 4.1 and of the theorem. R2 re-ran the float certificate and checked Lemmas 3.2 and 3.3 against the arXiv text of [GM]. Its verdict was *correct with minor fixable issues*. There were

eight issues, all addressed: clamping to the caps; a numerical slip; fixing θ before splitting H ; the move of the Vinogradov–Korobov bound to $\operatorname{Re} s = \frac{1}{2}$; log powers in Lemma 4.3; theorem numbers; reproducibility of a float run; and an optional exact rerun, done by X1. A fresh-eyes referee (R3) then read that note; its minor issues were applied. The phase II changes are:

- (1) the $\inf_k$ display of Lemma 3.3 as an input (E3k);
- (2) the box-edge argument: the old σ -edge step is invalid for E3k and is replaced by the strict slack;
- (3) the new certificate (§4.7) and Remark 5.1;
- (4) the corner numbers and the range of Lemma 4.1;
- (5) the parameter choice $\varepsilon \leq (\vartheta - 0.5435)/4$.

They were refereed by an independent agent (R4), who did not write them.

Referee R4. Verdict: *accept*, with no blocking or major issues. R4 found that Lemma 4.1 holds for $\tau \in [0.45, 0.4565]$ and that Theorem 1.1 follows; it also accepted Remark 5.1 and recommended keeping it as a remark. In detail, R4:

- read the $\inf_k$ display of [GM, Proposition 12.1] in the arXiv text and re-derived its exponent form;
- checked GA’s code diff against X1 (four changes, all minimal and exact);
- confirmed that the old σ -edge step fails for E3k and that the strict slack is necessary and sufficient;
- recomputed the corner numbers at $\tau = 0.4565$ and 0.456692 , and checked that Lemma 4.3 is independent of τ ;
- re-derived every inequality of the parameter choice;
- re-ran the certificate, with bit-identical leaf counts;
- re-verified every accepted leaf with the independent script `r4_verify.py`: 331 054 leaves at $\tau_{\text{hi}} = 913/2000$ and 586 580 at $\tau_{\text{hi}} = 0.456692$, with 0 failures (§4.7).

R4 raised four minor issues, all addressed in this version:

- m1: state E3k with the strict slack, cite `min0k_noclosed` as the primary certificate, and drop the closed form and the old edge step;
- m2: the script’s docstring omits the slack (noted in §4.7);
- m3: editorial consistency of the sanity runs, the old 0.544 remark (now removed), §6 and the certificate footnote;
- m4: cite the leaf-by-leaf re-verifier for reproducibility.

The note has *not* been refereed by humans. The main residual risks are:

- (i) The audit of the MT text (S2) relies on quotes obtained through a summarising web tool. An independent re-check (S3) agrees, and also used such a tool. In particular, the claim that [MT, Lemma 4.2] and the proof of [MT, Lemma 4.3] do not use $\theta > 0.55$ rests on that reading, together with our own re-derivation. R2 lists this as the one inherited caveat.
- (ii) Classical results are cited as standard, without theorem numbers (Remark 3.8).
- (iii) The new certificate was written by one agent (GA). Its code diff was reviewed by R4, and R4’s independent script re-verified its leaves. Like the earlier versions, it checks only the arithmetic of the box bounds, given the analytic inputs.
- (iv) The result now depends on the $\inf_k$ display of [GM, Proposition 12.1], not only on its closed form; the closed form alone stops at 0.5437644.

We have not searched the literature exhaustively for prior applications of [GM] to this problem.

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